Session 5 Lab Deliverables

Industrial Maintenance Work Order Modernization

Part A – Sequence Diagram Content

1. System Sequence Diagram (SSD) – Create Work Order

Lifelines:

- Requester

- Work Order System (Black Box)

Happy Path Messages (in order):

1. submitWorkOrder(asset_id, priority, description)

2. validateMandatoryFields()

3. generateWorkOrderId()

4. calculateSlaTarget(priority)

5. saveWorkOrder(details, sla_target)

6. notifyTechnician(work_order_id)

7. confirmation(work_order_id, sla_target)

Exception Flow (alt/else):

If mandatory fields are missing → return validationError(message)

Assumptions:

1. Priority determines SLA target.

2. SLA target is retrieved from SLAs table.

3. Work order cannot be created without asset_id.

4. Notification occurs after successful persistence.

5. System assigns unique work_order_id.

Part B – Data Dictionary Excerpt

Table: WorkOrders

Column	Definition	Type	Null?	Allowed/Rule	Example
work_order_id (PK)	Unique identifier for work order	INT	No	Must be unique	10045
asset_id (FK)	Reference to asset	INT	No	Must exist in Assets table	2001
technician_id (FK)	Assigned technician	INT	Yes	Must exist in Technicians	301
priority	Urgency level for SLA selection	ENUM	No	P1,P2,P3,P4	P2
status	Current state of work order	ENUM	No	Open, In Progress, Closed	Open
created_at	Timestamp of creation	DATETIME	No	Must be <= closed_at	2026-01-10 14:22
started_at	Timestamp when repair started	DATETIME	Yes	Must be >= created_at	2026-01-10 15:00
closed_at	Timestamp of closure	DATETIME	Yes	Must be >= started_at	2026-01-10 18:00
sla_target_minutes	Allowed resolution time	INT	No	>=0	240
downtime_minutes	Total downtime caused	INT	Yes	>=0	75

Table: SLAs

Column	Definition	Type	Null?	Allowed/Rule	Example
sla_id (PK)	Unique SLA identifier	INT	No	Must be unique	1
priority	Priority linked to SLA	ENUM	No	P1,P2,P3,P4	P2
target_minutes	Resolution time target	INT	No	>=0	240
breach_threshold_minutes	Time considered breach	INT	No	>= target_minutes	300
effective_from	SLA start date	DATE	No	Valid date	2026-01-01
effective_to	SLA end date	DATE	Yes	>= effective_from	2026-12-31

Automated Data Quality Tests

1. work_order_id must be unique.

2. created_at must be less than or equal to closed_at.

3. priority must be one of (P1,P2,P3,P4).

Part C – KPI Specification Sheet

KPI	Business Question	Formula	Grain	Segments	Target
SLA Compliance %	Are work orders resolved within SLA?	Closed within sla_target_minutes / total closed WOs	Per Work Order	By priority, location	>=90%
Backlog Count	How many work orders are	Count where status != 'Closed'	Per Day	By age bucket	<50

	open?				
MTTR	Average repair time?	AVG(closed_at - started_at)	Per Work Order	By asset, technician	<4 hours
Downtime Minutes	Total downtime caused?	SUM(downtime_minutes)	Per Asset	By location	Reduce monthly
SLA Breach Count	How many WOs breached SLA?	Count where resolution_time > sla_target_minutes	Per Month	By priority	Minimize
Technician Utilization	How efficiently technicians are used?	Total repair hours / available hours	Per Technician	By team	>=80%

Pseudo-SQL Examples

1. SLA Compliance %

```
SELECT
  (COUNT(CASE WHEN TIMESTAMPDIFF(MINUTE, created_at, closed_at) <=
sla_target_minutes THEN 1 END) * 100.0
  / COUNT(*)) AS sla_compliance_percentage
FROM WorkOrders
WHERE status = 'Closed';
```

2. Backlog Count

```
SELECT COUNT(*) AS backlog_count
FROM WorkOrders
WHERE status != 'Closed';
```

KPI Not Computable

MTBF (Mean Time Between Failures) cannot be computed because operating hours data per asset is not available.